CHATCO System Test Cases

All phases: 214 test cases covering the commuter and conductor mobile applications.

Test data: password **Chatco@123**, contact number **09171234567**, and a clear photo of a valid ID (JPG, PNG, or WebP, under 5 MB).

Status: P = Passed, F = Failed. Cases marked with a note number are expected to fail until that issue is fixed (see the last page).

Phase	Roles	Modules	Test Cases	Case IDs
Phase 1: Commuter Mobile Authentication and Onboarding	Public, Commuter	First-Launch Walkthrough, Login, Signup, OTP Verification, Platform-Scoped Tokens	**18**	001 to 018
Phase 2: Commuter Live Radar and Smart Hailing	Commuter	Full-Bleed CARTO Map, Tile Failover, GPS Beacon, 1km Radius Guard, Hail Request & Expiry	**15**	019 to 033
Phase 3: Commuter Cashless GCash Checkout and Ride History	Commuter	Camera QR Viewfinder, Manual Token Fallback, Concession Fare Discounts, GCash Webview Checkout, Ride History	**14**	034 to 047
Phase 4: Commuter Paper Receipt Scanner and Rewards Wallet	Commuter	Physical Receipt Scanner, 10-Ride Progress Cycle, Free Ride Vouchers, Conductor Activation Code, Announcements	**16**	048 to 063
Phase 5: Commuter Safety, Lost and Found, and Ride Feedback	Commuter	30-min Ride Share URL, Emergency Double-Tap SOS, Lost Belongings Catalog & Claim Proof, Crew Rating	**34**	064 to 097
Phase 6: Commuter Profile and Conductor Shift Initialization	Commuter, Conductor	Commuter Profile & Security, Conductor Mobile Login, Unit Verification, Paired Driver Verification	**24**	098 to 121
Phase 7: Conductor Operating Device, GPS Broadcast, and Capacity	Conductor	Operating Device Ownership, View-Only Mode, Shift Device Handoff, GPS Tracking, Capacity Toggle	**18**	122 to 139
Phase 8: Conductor Offline Ticketing, GCash QR, Hails, and Remittance	Conductor	Offline Fare Calculator, Concession Discounts, GCash Fare QR, Voucher Verification, Hail Dispatch, Shift Closeout	**51**	140 to 190
Phase 9: Mobile Hardware, Cross-Platform Interoperability, and Reliability	Commuter, Conductor, System	End-to-End QR Pay & Hail Sync, Concurrent Sessions, Camera Torch & Permissions, Background GPS, Airplane Mode Sync	**24**	191 to 214
Total			**214**	

Phase 1: Commuter Mobile Authentication and Onboarding

Roles: Public, Commuter

A. First-Launch Onboarding Guide

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-AUTH-001	First-launch onboarding walkthrough appears	Fresh installation / storage cleared (**@chatco_commuter_onboarding_completed** is null)	1. Launch the mobile commuter application	The 4-step onboarding guide screen appears immediately before login. Shows Step 1 badge: "REAL-TIME RADAR", CARTO dark visual, and "Continue" button.	
TC-COMM-AUTH-002	Step through the 4 onboarding slides	Onboarding screen open at Step 1	1. Read Step 1 and tap "Continue" 2. Read Step 2 (QR Pay & Vouchers) 3. Read Step 3 (Safety & SOS) 4. Read Step 4 (Lost & Found)	Progress dots update smoothly from 1 to 4. Each step renders distinct accent colors, icons, and 3 feature highlight cards. Step 4 button changes to "Get Started".	
TC-COMM-AUTH-003	Complete onboarding and verify persistence	On Step 4 of Onboarding	1. Tap "Get Started" 2. Force close the app 3. Reopen the app	App saves key **@chatco_commuter_onboarding_completed = true**. Login screen is displayed immediately. Onboarding does not appear again.	
TC-COMM-AUTH-004	Skip onboarding	On Step 1 of Onboarding	1. Tap "Skip" in the top-right header	Onboarding exits immediately, flags completion in AsyncStorage, and transitions to Login.	
TC-COMM-AUTH-005	Navigate backwards in onboarding	On Step 3 of Onboarding	1. Tap "Back"	Transitions smoothly to Step 2 with preserved pagination state and highlights.	

B. Commuter Mobile Login

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-AUTH-006	Valid Commuter login with Email	Approved commuter account exists	1. Enter **juan.commuter@gmail.com** 2. Enter **Chatco@123** 3. Tap "Sign In"	Authenticates via **/api/v1/mobile/commuter/login**, stores Sanctum token securely in **expo-secure-store**, redirects to full-bleed Dashboard map with greeting: "Good morning, Juan!".	
TC-COMM-AUTH-007	Valid Commuter login with Username	Approved commuter account exists	1. Enter **juan.commuter** 2. Enter **Chatco@123** 3. Tap "Sign In"	Login succeeds and loads commuter profile seamlessly.	
TC-COMM-AUTH-008	Login with incorrect password	Commuter account exists	1. Enter valid email and wrong password 2. Tap "Sign In"	Displays inline error banner: "Invalid credentials". Password field remains clearable.	
TC-COMM-AUTH-009	Login with unregistered email	Account does not exist	1. Enter **unregistered@gmail.com** 2. Tap "Sign In"	Displays error banner: "Invalid credentials". No session created.	
TC-COMM-AUTH-010	Login with blank credentials	On Login screen	1. Leave fields blank 2. Tap "Sign In"	Inline form validation blocks submission: "Please enter your email or username" and "Please enter your password".	
TC-COMM-AUTH-011	Commuter account pending admin approval	Account signed up but unverified by Admin	1. Enter pending commuter credentials 2. Tap "Sign In"	Error banner displays: "Your account is pending admin approval." Commuter is blocked from accessing the dashboard.	
TC-COMM-AUTH-012	Commuter account suspended	Admin marked commuter as suspended	1. Enter suspended commuter credentials 2. Tap "Sign In"	Displays alert modal: "This account has been suspended by administration. Reason: [Reason]."	

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-AUTH-013	Commuter password visibility toggle	Password typed in login	1. Type password 2. Tap eye icon	Password toggles between obscured dots (••••••••) and plain text.	

C. Commuter Mobile Registration

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-AUTH-014	Complete valid 4-step mobile registration	Unregistered email and contact number	1. Tap "Sign Up" 2. Step 1: Enter First Name, Surname, Birthday, Gender, Commuter Type (Student) 3. Step 2: Enter Email, 09171234567, capture/upload School ID 4. Step 3: Enter 6-digit email OTP 5. Step 4: Create Username and Password	Transitions to "Registration Submitted!" confirmation screen. Informs commuter account is pending admin verification.	
TC-COMM-AUTH-015	Registration invalid ID file format / size	Step 2 of Registration	1. Attempt to upload a PDF or file > 5MB	Error banner displays: "That file type isn't supported. Upload a JPG, PNG, or WebP image under 5MB."	
TC-COMM-AUTH-016	Registration OTP resend timer countdown	Step 3 of Registration	1. Tap "Resend Code" 2. Attempt to tap immediately again	60-second countdown timer initiates ("Resend available in 59s..."). Button is disabled until timer reaches 0.	

D. Platform-Scoped Session Isolation

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-AUTH-017	Platform-scoped session isolation	Commuter logged in on Web browser	1. Log in to Commuter Mobile app with same credentials 2. Refresh/navigate on Web browser	Both Web and Mobile sessions remain fully active concurrently. Mobile token carries **abilities: ["role:commuter", "platform:mobile"]**. Neither session drops.	
TC-COMM-AUTH-018	Commuter logout	Logged in on Commuter Mobile	1. Go to Profile 2. Tap "Log Out" 3. Confirm in modal	Revokes mobile Sanctum token from backend, purges secure storage, and returns to Login screen.	

Phase 2: Commuter Live Radar and Smart Hailing

Roles: Commuter

A. Full-Bleed Map & Live Tracking

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-HAIL-019	Full-bleed CARTO map initial load	Commuter logged in, on Home tab	1. Inspect dashboard screen layout	CARTO Dark basemap renders full-bleed (**StyleSheet.absoluteFill**). Floating frosted glass greeting sits top-left; avatar sits top-right; bottom sliding sheet sits at bottom. No static header bar cuts into the map.	
TC-COMM-HAIL-020	CARTO tile error automatic failover	CARTO API returns 401 or network blocked	1. Simulate tile error on CARTO endpoint	Error listener triggers automatic silent fallback to OpenStreetMap tiles. Map continues rendering without gray screen or crashes.	
TC-COMM-HAIL-021	Active jeepney markers on corridor	Shift active for Unit J-04	1. Observe corridor on map	Unit J-04 pin appears with vehicle icon, plate number badge, and route direction along McArthur Highway.	
TC-COMM-HAIL-022	GPS location permission granted	App opened first time	1. Grant "While using the app" permission	User's blue location beacon renders on map with pulsating accuracy ring. Status indicator displays "Live Tracking Active".	
TC-COMM-HAIL-023	GPS location permission denied	Location permission disabled	1. Deny location permission	Map defaults to route midpoint (Malolos/Bulacan). Status card displays yellow warning: "Location access denied. Enable location to see ETA and hail vehicles."	

B. 1km Radius Guard & Hailing Dispatch

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-HAIL-024	Commuter within 1km service radius	User within 800m of active Unit J-04	1. Open Dashboard sheet	Live ETA card turns emerald green: displays **~3 min**, **0.8 km away**, and plate **NDB-5821**. Circular "Hail Me" button illuminates orange (**#FF6D3A**) with glowing border. Subtext reads "HAIL ME".	
TC-COMM-HAIL-025	Commuter outside 1km service radius	User > 1km from nearest unit	1. Open Dashboard sheet	Status card shows: "Outside service radius. Move within 1km of a conductor unit to hail." Hail button turns disabled gray (**#374151**) with subtext "TOO FAR".	
TC-COMM-HAIL-026	Trigger valid "Hail Me" request	In 1km radius, Hail button active	1. Tap circular "Hail Me" button	Button initiates spinner -> turns pulsating red (**#EF4444**) with **X** icon. Subtext displays "CANCEL". 3-minute digital countdown begins. Conductor app receives incoming hail alert instantly.	
TC-COMM-HAIL-027	Audible and haptic hail feedback	Hail button tapped	1. Tap "Hail Me" on physical device	Device plays soft dual-tone chime and triggers haptic vibration pattern **[100, 50, 100]ms**.	
TC-COMM-HAIL-028	Conductor accepts hail	Hail is pending	1. Conductor taps "Accept"	Commuter sheet immediately updates with green banner: "Conductor Accepted! Unit J-04 is approaching your stop."	
TC-COMM-HAIL-029	Conductor rejects hail	Hail is pending	1. Conductor taps "Decline"	Commuter sheet displays notification: "Unit is currently at maximum capacity. Please hail the next approaching unit." Button resets to "HAIL ME".	

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-HAIL-030	Commuter cancels active hail	Hail is active (countdown running)	1. Tap red "Cancel" button	Hail request is cancelled via **/api/v1/mobile/commuter/hail/{id}**. Conductor pin clears. Button returns to orange "HAIL ME".	
TC-COMM-HAIL-031	Hail 3-minute auto-expiration	Hail active for 180 seconds	1. Let timer reach 00:00 without conductor action	Hail auto-expires on backend. App plays subtle reminder tone and resets button to "HAIL ME".	
TC-COMM-HAIL-032	Hail unit that is on break	Conductor has set status to "On Break"	1. Attempt to hail unit	System prevents hail: "Vehicle is currently on break. Nearest active unit will be hailed."	
TC-COMM-HAIL-033	Expand and collapse bottom sheet	On Dashboard	1. Tap drag handle / "Tools" bar	Sheet collapses down to 38px height with "TOOLS" label, giving unobstructed map view. Tapping again smoothly expands full tools.	

Phase 3: Commuter Cashless GCash Checkout and Ride History

Roles: Commuter

A. QR Scanner & GCash Checkout

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-PAY-034	Open GCash payment card	Logged in on Home tab	1. Tap blue "Pay with GCash" card in sliding sheet	Opens full-screen **QrScannerModal** with live viewfinder and targeting reticle.	
TC-COMM-PAY-035	Scan valid conductor vehicle QR code	Conductor generated GCash fare QR	1. Point camera at conductor QR	Viewfinder beeps and vibrates. Modal resolves crew details: Unit J-04, Conductor C. Dela Cruz, Fare Amount PHP 15.00. Prompts "Proceed to GCash Checkout".	
TC-COMM-PAY-036	Student concession discount applied	Commuter is verified Student (20% discount)	1. Scan fare QR for PHP 15.00 regular trip	Fare checkout displays regular PHP 15.00, discount -PHP 3.00, final payable PHP 12.00 with badge "Student 20% Applied".	
TC-COMM-PAY-037	Senior Citizen / PWD discount applied	Commuter is verified Senior / PWD	1. Scan fare QR for PHP 25.00 trip	Calculates 20% discount: PHP 20.00 final amount. Concession role noted in transaction metadata.	
TC-COMM-PAY-038	Camera permission denied, switch to manual token	Camera permission blocked in device settings	1. Tap "Pay with GCash" 2. Tap "Enter token manually"	Input field appears. Allows entering 6-character alphanumeric QR token (e.g. **K9X2P4**).	
TC-COMM-PAY-039	Paste invalid / non-existent QR token	Manual entry open	1. Enter **INVALID** 2. Tap "Resolve Token"	Inline error displays: "We don't recognise this QR code. Please verify with conductor."	
TC-COMM-PAY-040	Redirect to GCash Payment Gateway	Valid checkout generated	1. Tap "Continue to GCash"	Deep-links to GCash app or opens secure GCash gateway in browser webview with pre-filled reference and exact fare amount.	
TC-COMM-PAY-041	Successful GCash payment return	Commuter completed payment in GCash	1. Complete payment and return to app	App displays green checkmark: "Payment Successful! Receipt Reference: GC-9482104". Conductor terminal receives live update: "PAID".	
TC-COMM-PAY-042	Cancel on GCash gateway	On GCash webview	1. Tap "Cancel" or return back	Displays notice: "The payment was cancelled. Please try again or pay with cash."	
TC-COMM-PAY-043	QR already claimed by another commuter	Another passenger scanned same single QR	1. Scan the already-claimed QR	Error banner shows: "This payment has already been claimed by another commuter. Please ask conductor for a new QR."	
TC-COMM-PAY-044	Group ride payment checkout	Conductor generated group fare (3 pax)	1. Scan group fare QR	Checkout modal reflects: "Group Ride (3 Passengers)". Gross: PHP 45.00. One single GCash transaction covers entire group.	

B. Ride History & Past Receipts

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-PAY-045	View Payment History modal	Paid rides exist	1. Tap clock icon on GCash Pay card	Opens **PaymentHistoryModal** displaying list of past rides with date, unit, route, amount, and payment method badge (**GCASH** / **CASH**).	
TC-COMM-PAY-046	Filter payment history	History modal open	1. Filter by "This Week", then "This Month"	Records filter instantly with total spent recalculating dynamically.	
TC-COMM-PAY-047	Open trip feedback from payment history	Trip completed without feedback	1. Tap "Leave Feedback" on past trip row	Closes history modal and deep-links directly to Feedback screen with pre-filled crew details.	

Phase 4: Commuter Paper Receipt Scanner and Rewards Wallet

Roles: Commuter

A. Paper Receipt Scanner

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-REWARD-048	Open paper receipt scanner from Rewards	Commuter paid via cash, holds thermal paper receipt	1. Go to Rewards tab 2. Tap QR scanner button in top-right header	Opens **ReceiptScanModal** with camera targeting receipt barcode/QR.	
TC-COMM-REWARD-049	Scan valid paper cash receipt	Receipt contains valid 6-char token	1. Scan receipt QR	Chime sounds. Displays success banner: "Receipt Claimed! PHP 15.00 cash ride added to your rewards progress." Progress bar advances by 1 ride.	
TC-COMM-REWARD-050	Enter paper receipt token manually	Holding paper receipt	1. Tap "Enter 6-character receipt code" 2. Type token (e.g. **R7C2M8**) 3. Tap "Claim Receipt"	Successfully claims receipt. Increments commuter's ride cycle count.	
TC-COMM-REWARD-051	Claim already claimed receipt	Receipt was already claimed	1. Scan same receipt a second time	Error displays: "You have already claimed this receipt." Ride count does not duplicate.	
TC-COMM-REWARD-052	Claim receipt claimed by different commuter	Commuter B scans Commuter A's receipt	1. Scan the used receipt	Error displays: "This receipt has already been claimed by another commuter."	
TC-COMM-REWARD-053	Claim expired paper receipt	Receipt date > claim window (e.g. 7 days)	1. Scan expired receipt	Error displays: "Receipt Expired. Cash receipts must be claimed within 7 days of issuance."	

B. Rewards Wallet & Vouchers

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-REWARD-054	10-Ride reward progress cycle bar	Commuter has 7 completed cashless/cash claimed rides	1. View Free Ride Progress card on Rewards tab	Displays "7 / 10 rides completed", progress bar at 70%, and subtitle "3 more rides until your next free ride voucher!".	
TC-COMM-REWARD-055	Unlock Free Ride Voucher on 10th ride	Commuter at 9 rides, claims 10th ride	1. Complete 10th ride payment or receipt claim	Celebration modal triggers: "■ Free Ride Unlocked!". New voucher appears in "Available Vouchers" section with 30-day validity. Cycle counter resets to 0/10.	
TC-COMM-REWARD-056	View available vouchers in wallet	Commuter holds 1 active voucher	1. Open Rewards tab	Voucher ticket card displays: "100% OFF Modern Jeepney Ride", expiry date, and "Use Voucher" action button.	
TC-COMM-REWARD-057	Open Conductor Activation Code modal	Voucher is available	1. Tap "Use Voucher"	Modal reveals large scannable Voucher QR code and human-readable 6-character code (e.g. **VCH-7X9**) to present to conductor.	
TC-COMM-REWARD-058	Copy voucher code to clipboard	Activation modal open	1. Tap "Copy Code" button	Copies code to system clipboard and displays toast: "Voucher code copied!".	
TC-COMM-REWARD-059	Redeem voucher for GCash checkout	Voucher available, scanning fare QR	1. Scan conductor's GCash QR 2. Select "Apply Free Ride Voucher"	Payable amount updates to PHP 0.00: "Voucher covers this ride in full -- no GCash payment needed." Confirms ride immediately.	
TC-COMM-REWARD-060	View Archived / Used Vouchers	Past vouchers redeemed	1. Scroll to "Voucher History"	Lists past vouchers with **USED** status badge and timestamp of redemption.	

C. Announcements Feed

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-REWARD-061	Dynamic Announcements feed display	Admin published System & Promo announcements	1. Scroll to "Latest Updates" on Rewards tab	Announcements display with categorized badges (**PROMO** ■, **SAFETY** ■■, **ADVISORY** ■) and relative timestamp. Unread items have orange indicator dot.	
TC-COMM-REWARD-062	View full announcement modal & mark as read	Unread announcement exists	1. Tap announcement card	Opens modal with complete announcement message. Orange unread dot clears and layout bottom badge decrements by 1.	
TC-COMM-REWARD-063	Mark all announcements as read	3 unread announcements exist	1. Tap "Mark all as read" button	All unread dots disappear. Bottom nav Rewards badge count clears to 0.	

Phase 5: Commuter Safety, Lost and Found, and Ride Feedback

Roles: Commuter

A. Safety Shield & Emergency Response

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-SAFE-064	Generate 30-minute Share Ride tracking link	Commuter on active transit, GPS enabled	1. Tap "Share Ride" button on Dashboard	Creates live tracking link via **/api/v1/mobile/commuter/share-ride**. Modal displays URL **https://chatco.ph/track/{token}** with 30:00 countdown timer.	
TC-COMM-SAFE-065	Copy and share tracking link	Share Ride modal open	1. Tap "Copy Link" or "Share Link"	Copies URL to clipboard or opens native iOS/Android share sheet (SMS, WhatsApp, Messenger).	
TC-COMM-SAFE-066	Continuous GPS broadcast during ride share	Sharing is active	1. Move along corridor for 5 minutes	App broadcasts updated coordinates every 10 seconds to backend. Family viewing link sees live pin moving on CARTO map.	
TC-COMM-SAFE-067	Manually terminate ride share	Sharing is active	1. Open Share modal 2. Tap "Stop Sharing"	Terminates tracking session. Public link transitions to "Ride Completed / Tracking Ended".	
TC-COMM-SAFE-068	Share link automatic expiration	Link generated 30 minutes ago	1. Let timer expire	Link expires automatically. Anyone accessing URL sees "This tracking link has expired."	
TC-COMM-SAFE-069	Emergency SOS double-tap protection	On Dashboard	1. Tap "Emergency SOS" once	Button changes text to red: "Tap SOS again". SOS is NOT triggered yet (prevents accidental pocket taps).	
TC-COMM-SAFE-070	Trigger confirmed Emergency SOS	"Tap SOS again" is displayed	1. Tap button a second time within 1.5 seconds	Modal triggers: "EMERGENCY SOS BROADCAST ACTIVE". Coordinates sent to Admin Dispatch, emergency alert email dispatched, siren haptic vibrates device.	
TC-COMM-SAFE-071	Emergency SOS rate limiting	SOS was just triggered	1. Attempt to send another SOS within 60 seconds	Toast warns: "You have already sent an SOS recently. Please wait before trying again."	
TC-COMM-SAFE-072	One-tap PNP Police 911 hotline dial	SOS modal open	1. Tap "Dial PNP 911" button	Native phone dialer opens with **911** pre-filled ready for one-tap calling.	
TC-COMM-SAFE-073	One-tap BFP Fire / Rescue 160 hotline dial	SOS modal open	1. Tap "Dial BFP 160" button	Native phone dialer opens with **160** pre-filled.	

B. Centralized Lost and Found Recovery

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-LOST-074	View Lost & Found catalog	Commuter on Lost & Found tab	1. Inspect screen layout	Segmented control displays 3 tabs: "All Items", "Watchlist", "Claims". Item cards show photo thumbnail, item title, category badge, found date, and unit plate.	
TC-COMM-LOST-075	Search lost items by keyword	Items exist in catalog	1. Type "wallet" in search input	Catalog filters instantly to matching items across title, description, and unit notes.	
TC-COMM-LOST-076	Filter items by category chip	On "All Items" tab	1. Tap "Gadgets & Electronics" chip	Displays only gadgets (smartphones, earphones, powerbanks). Non-matching categories hidden.	
TC-COMM-LOST-077	Filter items by date lost	On "All Items" tab	1. Pick specific date from date picker	Catalog shows items recovered on that specific date.	

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-LOST-078	View complete item detail modal	Item card visible	1. Tap item card	Opens modal displaying high-resolution image, full description, recovered unit (**J-04**), driver/conductor names, and status badge (**AVAILABLE**).	
TC-COMM-LOST-079	Add item to personal watchlist	On item card	1. Tap bookmark icon	Bookmark fills blue. Item is pinned to commuter's "Watchlist" tab.	
TC-COMM-LOST-080	Remove item from watchlist	On Watchlist tab	1. Tap bookmark icon on watched item	Item is removed from Watchlist tab immediately.	
TC-COMM-LOST-081	File claim for lost item	Available item open	1. Tap "Claim Item" 2. Fill in unique identifying marks/proof: "Black leather wallet with PhilHealth ID inside" 3. Enter contact: **09171234567** 4. Submit	Claim is recorded via **/api/v1/lost-found/{id}/claim**. Status moves to **PENDING** ("Waiting for staff review"). Item moves to "Claims" tab.	
TC-COMM-LOST-082	File claim with blank proof	Claim form open	1. Leave proof description blank 2. Tap Submit	Validation alert displays: "Please describe identifying details or unique marks." Submission blocked.	
TC-COMM-LOST-083	Admin approves claim -- commuter notification	Claim reviewed and approved by staff	1. Open app or check Lost & Found tab	Nav badge increments. Claim status updates to green **APPROVED / READY** with instruction: "Approved! Bring valid ID to terminal office to claim."	
TC-COMM-LOST-084	Admin rejects claim	Staff marked claim invalid	1. Open Claims tab	Status updates to red **REJECTED** with staff rejection rationale (e.g. "Identifying details do not match item contents").	
TC-COMM-LOST-085	Cancel pending claim	Claim is in **PENDING** status	1. Tap "Cancel Claim" 2. Confirm in dialog	Claim is withdrawn. Item returns to available catalog for other claimants.	
TC-COMM-LOST-086	Maximum 3 rejected claims threshold	Commuter had 3 rejected claims on item	1. Attempt to file 4th claim on same item	System blocks submission: "You have reached the maximum number of rejected claims for this item."	
TC-COMM-LOST-087	Item released by staff	Item physically handed to commuter	1. Open Claims tab	Claim displays final blue badge: **RELEASED** with timestamp and releasing officer details.	

C. Crew Ride Feedback & Rating

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-FEED-088	Open feedback unit scanner	Commuter on Feedback tab	1. View Feedback screen	Opens in-page unit QR scanner with manual token entry option below.	
TC-COMM-FEED-089	Scan vehicle feedback QR code	Unit J-04 feedback QR	1. Scan unit QR code	Viewfinder resolves crew: Unit J-04, Driver R. Santos, Conductor C. Dela Cruz, Calumpit - Meycauayan Line. Opens dual rating form.	
TC-COMM-FEED-090	Rate Driver with 5-star rating	Crew rating form open	1. Tap 5th star on Driver section	All 5 stars illuminate gold (**#FBBF24**). Rating label shows "Excellent". Positive tag chips expand.	
TC-COMM-FEED-091	Select Driver feedback tag chips	5-star rating selected	1. Tap "Safe Driving" and "Smooth Ride"	Tag chips highlight blue (**#1A5FB4**). Selected tags added to submission payload.	
TC-COMM-FEED-092	Rate Driver with 1-star (negative feedback)	Rating form open	1. Tap 1st star on Driver section	Star illuminates. Label shows "Terrible". Negative tags expand: "Reckless Driving", "Loud Music", "Overspeeding".	

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-FEED-093	Rate Conductor section	Conductor section visible	1. Tap 4 stars for Conductor 2. Select tags "Polite" and "Honest Change"	Conductor rating and tags are selected. Both crew sections are now satisfied.	
TC-COMM-FEED-094	Submit feedback with optional comment	Driver and Conductor rated	1. Enter comment: "Driver was very careful during rain." 2. Tap "Submit Feedback"	Feedback submitted via **/api/v1/mobile/commuter/feedback**. Green success screen appears: "Thank You! Your feedback has been recorded."	
TC-COMM-FEED-095	Auto-redirect to Dashboard after feedback	Feedback submitted	1. Observe screen for 3 seconds	3-second countdown completes ("Redirecting in 3... 2... 1..."). Transitions cleanly back to Dashboard.	
TC-COMM-FEED-096	Submit without rating conductor	Driver rated, conductor untouched	1. Tap "Submit Feedback"	Alert blocks submission: "Please rate both the driver and the conductor before submitting."	
TC-COMM-FEED-097	Prevent duplicate feedback for same ride	Feedback already submitted for shift	1. Scan same unit QR again	Error banner shows: "You have already left feedback for this ride today. Thank you for helping us improve!"	

Phase 6: Commuter Profile and Conductor Shift Initialization

Roles: Commuter, Conductor

A. Commuter Profile & Security

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COMM-PROF-098	View Profile details	Logged in as commuter	1. Open Profile tab	Displays circular avatar with initials, full name, **@username**, verification badge, commuter type (Student), and read-only personal fields (Email, Birthdate).	
TC-COMM-PROF-099	Edit contact number	Profile tab open	1. Tap "Edit Profile" 2. Change contact to **09189876543** 3. Tap "Save Changes"	Updates contact via PUT **/api/v1/mobile/commuter/profile**. Toast confirms: "Profile updated successfully!". Contact number updates across all modals.	
TC-COMM-PROF-100	Edit with invalid contact number format	Edit profile mode active	1. Enter **0917** 2. Tap "Save Changes"	Validation error displays: "Enter an 11-digit mobile number starting with 09 (e.g. 09171234567)."	
TC-COMM-PROF-101	Request 2-Step Password Reset Code	Profile tab open	1. Tap "Change Password" 2. Enter current password and new valid password 3. Tap "Continue"	System sends 6-digit OTP code to registered email address. Modal transitions to Code Verification step.	
TC-COMM-PROF-102	Change password with wrong current password	In Change Password modal	1. Enter wrong current password 2. Enter new password 3. Tap "Continue"	Error banner shows: "The provided current password is incorrect." Code is not sent.	
TC-COMM-PROF-103	Change password confirmation mismatch	In Change Password modal	1. Enter two different new passwords	Error banner shows: "The new password confirmation does not match."	
TC-COMM-PROF-104	Change password weak password rules	In Change Password modal	1. Enter **chatco12** (no uppercase or symbol)	Checklist highlights missing requirements: (8+ chars, uppercase, number, symbol).	
TC-COMM-PROF-105	Enter wrong email verification OTP	Code verification step	1. Enter **999999**	Error banner shows: "That code is incorrect. Please check your email."	
TC-COMM-PROF-106	Enter correct OTP and confirm password reset	Valid OTP received	1. Enter valid 6-digit code 2. Tap "Confirm Change"	Password updated successfully via **/api/v1/mobile/commuter/change-password/confirm**. Modal closes and success banner appears. Old password no longer works.	
TC-COMM-PROF-107	Replay App Guide & Walkthrough from Profile	Profile tab open	1. Tap "App Features & Guide" card	Launches the 4-step first-launch Onboarding Screen walkthrough. Commuter can review all features and return to profile.	
TC-COMM-PROF-108	Replay App Guide from Top Header Bar	On any secondary tab	1. Tap **?** help circle icon in top header	Launches the interactive walkthrough guide immediately.	
TC-COMM-PROF-109	Logout confirmation modal	Profile tab open	1. Tap "Log Out" 2. Tap "Cancel" 3. Tap "Log Out" again and confirm	Tap 1 cancels and remains in profile. Tap 2 logs out, purges tokens, and returns to Login.	

B. Conductor Mobile Login

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COND-AUTH-110	Valid Conductor login	Conductor account created by admin	1. Launch **chatco-mobile** 2. Enter **c.delacruz** 3. Enter **Chatco@123** 4. Tap "Sign In"	Authenticates via **/api/v1/mobile/conductor/login**, saves Sanctum token with **abilities: ["role:conductor", "platform:mobile"]**, redirects to Unit Verification screen.	
TC-COND-AUTH-111	View available vehicles and drivers	Conductor logged in, no active shift	1. Inspect Unit Verification screen	Available units (**J-04**, **J-08**) and active drivers are listed in dropdown selectors.	
TC-COND-AUTH-112	Start shift with valid unit and driver	Unit J-04 and Driver R. Santos available	1. Select Unit **J-04** 2. Select Driver **R. Santos** 3. Review Shift Setup summary 4. Tap "Confirm Shift Setup"	Shift starts successfully. Operating device token registered. Conductor Dashboard opens with route Calumpit - Meycauayan.	
TC-COND-AUTH-113	Start shift without selecting unit or driver	Unit Verification open	1. Leave selectors blank 2. Tap "Confirm Shift Setup"	Validation error blocks start: "Vehicle and driver selection are required to start a shift."	
TC-COND-AUTH-114	Unit already on active shift by another conductor	Unit J-04 active on another shift	1. Attempt to start shift on Unit J-04	Error banner: "Vehicle is already assigned to an active shift."	
TC-COND-AUTH-115	Driver already on active shift	Driver R. Santos active on another unit	1. Attempt to select Driver R. Santos	Error banner: "Driver is already on an active shift."	

C. Unit Verification & Driver Pairing

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COND-AUTH-116	Conductor already has an open shift	Conductor has active shift from earlier	1. Log in from mobile app	Automatically bypasses Unit Verification and re-hydrates active shift dashboard directly.	
TC-COND-AUTH-117	Unit set to Maintenance mode by Admin	Admin set Unit J-04 to Maintenance	1. Select Unit J-04	Error banner: "This vehicle is currently under maintenance and cannot be operated."	
TC-COND-AUTH-118	Start shift during Maintenance Mode	Admin activated system Maintenance Mode	1. Attempt to start shift	Banner blocks start: "CHATCO is under maintenance right now. Shifts cannot be started until complete."	
TC-COND-AUTH-119	Password visibility toggle	On conductor login	1. Type password 2. Tap eye toggle	Password toggles between obscured dots and plain text.	
TC-COND-AUTH-120	Deactivated conductor account login	Admin disabled conductor account	1. Enter credentials for disabled conductor	Error displays: "This conductor account has been deactivated. Please contact admin."	
TC-COND-AUTH-121	Conductor session isolation	Conductor logged in on Web portal	1. Log in to Conductor Mobile app with same account	Mobile session establishes independently with mobile platform scope. Both can query shift data without dropouts.	

Phase 7: Conductor Operating Device, GPS Broadcast, and Capacity

Roles: Conductor

A. Operating Device Ownership & Handoff

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COND-DEV-122	Shift opened on second device (Device B)	Shift active on Device A (owns shift)	1. Log in on Device B with same conductor credentials 2. Open Dashboard	Device B detects Device A owns the shift. Device B renders in **View-Only Mode** with banner: "Shift active on another device. Choose operating device to take over."	
TC-COND-DEV-123	Collect fare on view-only device blocked	Device B is in view-only mode	1. Attempt to record fare or generate QR on Device B	Action is blocked: "This shift is active on another device. Sync and release that device before collecting fares here."	
TC-COND-DEV-124	Release shift for handoff on Device A	Shift active on Device A	1. On Device A, open Settings / Shift Options 2. Tap "Release for Handoff" 3. Confirm	Device A syncs all local transactions and releases ownership. Displays: "Shift is ready to be claimed by another device."	
TC-COND-DEV-125	Claim released shift on Device B	Device A released shift	1. On Device B, tap "Claim Operating Device"	Device B acquires shift ownership token. Full fare collection and controls unlock immediately.	
TC-COND-DEV-126	Claim while Device A has not released	Device A still holds ownership	1. On Device B, attempt to claim shift	Error displays: "Another device still owns this shift. Release it there before claiming this device."	
TC-COND-DEV-127	Block handoff when offline cash is unsynced	Device A recorded fares while offline	1. Attempt to tap "Release for Handoff" on Device A while offline	Warning blocks handoff: "Offline cash fares are still waiting to sync. Reconnect device to sync before handoff."	
TC-COND-DEV-128	Emergency shift recovery by Admin	Device A lost or damaged during shift	1. Admin triggers "Device Recovery" in admin panel 2. Conductor on Device B claims shift	Device B successfully claims shift. Old Device A ownership token is permanently invalidated.	
TC-COND-DEV-129	Old device attempts to operate after admin recovery	Device A recovered by admin	1. Reconnect Device A and attempt fare action	Error displays: "An administrator released this device from the shift. Please log in again."	

B. Live Status, GPS Broadcast & Capacity

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COND-STAT-130	Real-time GPS broadcasting	Shift active, location permission granted	1. Device moves along McArthur Highway corridor	Transmits coordinates every 5 seconds to **/api/v1/mobile/conductor/location**. Unit marker moves accurately on Admin and Commuter maps.	
TC-COND-STAT-131	Location permission denied warning	Shift active, GPS disabled	1. Disable location in phone settings	Dashboard displays high-priority GPS warning banner: "GPS signal lost. Enable location for live radar tracking." Admin map marks unit as "Awaiting GPS".	
TC-COND-STAT-132	Toggle passenger capacity to "Standing"	Shift active, seats full	1. Tap capacity selector -> Select "Standing"	Capacity status updates instantly to "Standing". Commuter app updates unit badge to show standing room available.	
TC-COND-STAT-133	Toggle passenger capacity to "Full"	Shift active	1. Tap capacity selector -> Select "Full"	Unit marks as "Full". Commuters cannot hail unit while full.	
TC-COND-STAT-134	Toggle passenger capacity to "Available"	Passengers alight	1. Tap capacity selector -> Select "Available"	Capacity resets to available with seat count visible to commuters.	

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COND-STAT-135	Start shift break duty	Shift active	1. Use "Slide to Start Break" slider 2. Confirm break start	Conductor status updates to "Conductor on Break". Unit hides from commuter hail radar. Admin monitoring reflects "On Break".	
TC-COND-STAT-136	Resume shift duty from break	Conductor is on break	1. Use "Slide to Resume Duty" slider	Status restores to "Conductor on Duty". Unit reappears on commuter hail radar.	
TC-COND-STAT-137	Network loss offline indicator banner	Shift active	1. Enable Airplane mode on phone	Amber top banner appears: "Offline Mode Active. Cash fares will be stored locally and synced when online."	
TC-COND-STAT-138	Network restoration and auto-sync	Device was offline with banner	1. Disable Airplane mode	Banner transitions to green: "Online. Syncing offline transactions..." then disappears once synced.	
TC-COND-STAT-139	Battery saver / background tracking	Device locked during transit	1. Lock screen while vehicle moves	Background geolocation task (**expo-task-manager**) maintains periodic coordinate pings along the route.	

Phase 8: Conductor Offline Ticketing, GCash QR, Hails, and Remittance

Roles: Conductor

A. Offline-First Fare Calculator & Ticketing

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COND-FARE-140	Fare calculation by origin and destination	Shift active	1. Open Fare Calculator 2. Select Origin: "Calumpit Junction" 3. Select Destination: "Malolos Crossing"	Displays exact fare according to published fare matrix (e.g. PHP 20.00 regular, PHP 16.00 discounted).	
TC-COND-FARE-141	Search stop by landmark or barangay	Fare calculator open	1. Type "Capitol" in stop search bar	Dropdown filters instantly to "Bulacan Provincial Capitol" stop.	
TC-COND-FARE-142	Apply Student concession fare	Origin and Destination selected	1. Select passenger type: "Student"	Fares recalculate to 20% concession rate (PHP 16.00 instead of PHP 20.00).	
TC-COND-FARE-143	Apply Senior Citizen / PWD concession fare	Origin and Destination selected	1. Select "Senior Citizen" or "PWD"	Fares recalculate to 20% concession rate. Concession tag attached to receipt.	
TC-COND-FARE-144	Record single cash fare payment	Fare calculated (PHP 20.00)	1. Tap "Cash" 2. Tap "Confirm Payment"	Records cash transaction. Displays receipt summary with 6-character claim token (e.g. **T7K9P2**). Increments shift cash total by PHP 20.00.	
TC-COND-FARE-145	Group cash fare calculation (multiple passengers)	Fare calculator open	1. Select "Multiple - Enter quantities" 2. Enter 2 Regular and 2 Student 3. Calculate	Calculates: (2 x PHP 20) + (2 x PHP 16) = PHP 72.00 total. Records single group cash ticket.	
TC-COND-FARE-146	Record cash fare while completely offline	Device in Airplane mode	1. Calculate fare 2. Select "Cash" 3. Tap "Confirm Payment"	Saves transaction into offline queue in local storage (**@chatco_conductor_offline_fares**). Generates local offline receipt token. Shifts cash total increments.	
TC-COND-FARE-147	Background sync of offline cash queue	3 cash fares stored offline	1. Reconnect phone to WiFi/mobile data	Background queue processor POSTs transactions to **/api/v1/mobile/conductor/fares/sync-batch**. Replaces offline IDs with permanent server IDs without duplicates.	
TC-COND-FARE-148	Prevent duplicate submission on double-tap	Fare confirmation button	1. Rapidly double-tap "Confirm Payment"	Button debounces immediately on first tap. Only one transaction is generated.	
TC-COND-FARE-149	View shift transaction history	Fares recorded during shift	1. Open "Transactions History" tab	Chronological list of all fares collected with timestamp, passenger count, fare amount, and payment method (**CASH**, **GCASH**, **VOUCHER**).	
TC-COND-FARE-150	Filter shift transactions	History open	1. Filter by "Cash Only", then "GCash Only"	List filters dynamically, displaying subtotal for filtered category.	
TC-COND-FARE-151	Thermal Bluetooth receipt printer pairing	Bluetooth enabled, portable printer on	1. Tap "Connect Printer" 2. Select paired printer	Connects successfully. Prints test receipt: "CHATCO Modern Transit - Calumpit Line".	
TC-COND-FARE-152	Print paper cash receipt	Fare confirmed, printer connected	1. Tap "Print Receipt"	Printer outputs paper ticket: Bus Unit, Conductor Name, Origin, Destination, Amount, Barcode, and 6-char Claim Token.	
TC-COND-FARE-153	Reprint previous receipt	Commuter asks for receipt copy	1. Open Transactions History 2. Select last transaction 3. Tap "Reprint"	Prints identical ticket with label "DUPLICATE RECEIPT".	
TC-COND-FARE-154	Reverse / void accidental fare	Misentered cash transaction	1. Open transaction within 2 minutes 2. Tap "Void Fare" 3. Enter reason	Transaction marks as **VOIDED**. Deducts amount from shift cash tally and logs audit record.	

B. GCash Fare QR & Voucher Redemption

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COND-PAY-155	Generate dynamic GCash fare QR code	Fare calculated (PHP 20.00), device online	1. Select "GCash" 2. Tap "Generate QR Code"	Displays high-contrast QR code on conductor screen with amount PHP 20.00 and status: "Waiting for commuter scan…".	
TC-COND-PAY-156	Real-time payment confirmation on QR	Commuter scans and pays via GCash	1. Commuter completes checkout	Screen immediately updates with green checkmark and audio bell: "Payment Successful! PHP 20.00 received via GCash." Increments shift GCash total.	
TC-COND-PAY-157	Cancel active GCash QR	QR displayed on screen	1. Tap "Cancel Payment"	Cancels transaction on backend. QR is invalidated. Returns to fare calculator.	
TC-COND-PAY-158	Dynamic GCash QR expiration	QR displayed for > 3 minutes	1. Wait for timer expiration	Displays message: "This QR code has expired. Tap to generate a new QR."	
TC-COND-PAY-159	Block GCash QR generation while offline	Device has no internet connection	1. Attempt to generate GCash QR	Warning banner displays: "GCash requires an active internet connection. Please accept cash while offline."	
TC-COND-PAY-160	Redeem valid commuter Free Ride Voucher	Commuter presents voucher code **VCH-7X9**	1. Select "Apply Voucher (Free Ride)" 2. Enter code **VCH-7X9** 3. Tap "Verify & Apply"	Code verifies via backend: "Valid Voucher: 100% Free Ride". Fare records at PHP 0.00. Shifts voucher count increments by 1.	
TC-COND-PAY-161	Scan commuter voucher QR code	Commuter presents voucher QR on phone	1. Tap "Scan Voucher QR" 2. Aim camera at commuter phone	Scanner reads voucher token instantly and applies discount.	
TC-COND-PAY-162	Enter invalid or non-existent voucher code	Voucher screen open	1. Enter **VCH-000** 2. Tap "Verify & Apply"	Error displays: "Invalid voucher code. Please check code or ask commuter to re-open wallet."	
TC-COND-PAY-163	Enter already redeemed / used voucher	Voucher was used earlier	1. Enter used voucher code	Error displays: "This voucher has already been redeemed on [Date]."	
TC-COND-PAY-164	Enter expired voucher	Voucher validity ended	1. Enter expired voucher code	Error displays: "This voucher has expired."	
TC-COND-PAY-165	Block voucher redemption while offline	Device in Airplane mode	1. Attempt to apply voucher	Warning displays: "Voucher validation requires internet connection to prevent reuse. Use cash while offline."	
TC-COND-PAY-166	Group ride with 1 voucher applied	Group of 3 passengers, 1 voucher presented	1. Apply 1 voucher to group ride	System applies voucher to 1 seat (PHP 0.00). Remaining 2 seats prompt for cash/GCash payment.	

C. Real-Time Hail Management

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COND-HAIL-167	Receive incoming commuter hail alert	Shift active, unit within 1km of commuter	1. Commuter taps "Hail Me" on mobile	High-priority pop-over alert appears on conductor dashboard with loud audio chime and vibration. Shows: "Incoming Hail: [Stop Name], 1 Passenger, 450m away".	
TC-COND-HAIL-168	Accept commuter hail	Incoming hail alert visible	1. Tap "Accept Hail" button	Hail changes status to **ACCEPTED**. Commuter app receives instant confirmation. Route pin updates with passenger indicator.	
TC-COND-HAIL-169	Decline commuter hail	Unit full or approaching terminal	1. Tap "Decline" button	Hail marks as declined. Alert dismisses. Commuter app receives polite notice to hail next unit.	

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COND-HAIL-170	Commuter cancels hail before conductor acts	Hail alert visible	1. Commuter cancels on their phone	Conductor hail modal dismisses automatically with subtle notification: "Hail was cancelled by commuter."	
TC-COND-HAIL-171	Hail expires automatically	Conductor did not act for 3 minutes	1. Let hail alert time out	Alert dismisses automatically without blocking conductor screen.	
TC-COND-HAIL-172	Block hails while unit status is Full	Conductor marked capacity as "Full"	1. Commuter attempts to hail	Conductor does not receive hail alert. Backend filters full unit from dispatch radar.	
TC-COND-HAIL-173	Block hails while conductor is on break	Status is "On Break"	1. Commuter attempts to hail	Conductor does not receive alerts.	
TC-COND-HAIL-174	Multiple incoming hails queue	2 commuters hail along corridor	1. Receive 2 hails within 15 seconds	Displays queue counter ("2 Hails Pending"). Allows accepting or declining each in order.	

D. Conductor Incident SOS Logging

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COND-SAFE-175	Conductor triggers Emergency SOS	Shift active, vehicle emergency	1. Tap red "SOS" button on header 2. Confirm "Send Emergency SOS?" in modal	Broadcasts high-priority alert with live coordinates, unit number (**J-04**), driver and conductor details to Admin Dashboard and dispatch email.	
TC-COND-SAFE-176	Emergency SOS rate limiting	SOS was just triggered	1. Attempt to send another SOS within 60 seconds	Warning displays: "Too many requests. Please slow down. Emergency alert is already active."	
TC-COND-SAFE-177	Display emergency response hotlines	SOS modal open	1. View emergency dialog	Displays direct speed-dial buttons for local PNP Station and modern jeepney fleet emergency dispatch.	
TC-COND-SAFE-178	Admin acknowledges conductor SOS	Conductor SOS is active	1. Admin clicks "Acknowledge" on web	Conductor screen displays banner: "SOS Acknowledged by Admin Dispatch. Help is on the way."	
TC-COND-SAFE-179	Log vehicle breakdown incident	Mechanical issue occurred	1. Open Incidents menu 2. Select "Vehicle Breakdown" 3. Enter note: "Flat tire near Bocaue" 4. Submit	Marks vehicle as disabled. Notifies dispatch to reroute subsequent units.	
TC-COND-SAFE-180	Log accident report	Collision or passenger injury	1. Open Incidents 2. Select "Accident" 3. Submit with GPS location	Immediately flags shift as high emergency on admin monitoring.	

E. Remittance, Reconcile & Shift Closeout

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COND-REMIT-181	View End of Day Remittance report	Shift active with recorded fares	1. Open "End of Day" tab	Displays complete breakdown: Total Passengers, Cash In Hand (PHP X), GCash Settled (PHP Y), Free Ride Vouchers (Z), and Grand Total Revenue.	
TC-COND-REMIT-182	Block remittance submission with unsynced offline fares	Offline fares remain in queue	1. Attempt to submit remittance while offline or unsynced	Submission is blocked: "Some cash transactions are still waiting to sync. Reconnect to internet before submitting remittance."	

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-COND-REMIT-183	Submit End of Day remittance	All transactions synced online	1. Tap "Submit Shift Report to Admin" 2. Confirm remittance submission	Shift closes permanently. Status transitions to **For Cash Declaration**. Displays Remittance Reference Number.	
TC-COND-REMIT-184	Prevent fare collection after shift closeout	Shift has been remitted	1. Attempt to open Fare Calculator	Blocked with message: "The shift has already ended; no new transactions can be recorded."	
TC-COND-REMIT-185	View past remittance history	Conductor completed past shifts	1. Open "Remittance History"	Lists past shift records with date, unit, gross cash, GCash, declaration status (**Complete**, **Shortage**, **Overage**).	
TC-COND-REMIT-186	Admin declares matching cash (Exact)	Remittance submitted (PHP 500 expected cash)	1. Admin inputs PHP 500 cash in Admin Remittance Tracker	Shift status updates to **Complete**. Conductor history reflects "Exact / Balanced".	
TC-COND-REMIT-187	Admin declares cash shortage	Expected PHP 500, Admin counts PHP 450	1. Admin inputs PHP 450	Status updates to **Shortage (PHP 50)**. Conductor history displays shortage note.	
TC-COND-REMIT-188	Admin declares cash overage	Expected PHP 500, Admin counts PHP 520	1. Admin inputs PHP 520	Status updates to **Overage (PHP 20)**. Conductor history displays overage note.	
TC-COND-REMIT-189	Automatic shift expiration past max shift hours	Max shift duration set to 12 hours	1. Shift reaches 12 hours without conductor closeout	Backend cron auto-closes shift to prevent stale sessions. Marks shift as **Auto-Ended: For Cash Declaration**.	
TC-COND-REMIT-190	Conductor logout after shift closeout	Shift is closed	1. Open Settings 2. Tap "Log Out" 3. Confirm	Conductor session logs out cleanly. Terminal returns to Conductor Login screen.	

Phase 9: Mobile Hardware, Cross-Platform Interoperability, and Reliability

Roles: Commuter, Conductor, System

A. Cross-Platform Interoperability

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-INT-191	Conductor QR to Commuter GCash Checkout end-to-end	Conductor on Unit J-04, Commuter on mobile	1. Conductor generates GCash QR for PHP 20 2. Commuter scans QR with mobile camera 3. Commuter pays via GCash	Conductor screen automatically flips to "PAID" via WebSocket/polling within 2 seconds. Commuter receives digital receipt.	
TC-INT-192	Commuter Hail to Conductor Terminal end-to-end	Commuter within 800m of Unit J-04	1. Commuter taps "Hail Me" 2. Conductor hears alert and taps "Accept"	Commuter screen updates to "Conductor Accepted". Conductor map plots pickup marker.	
TC-INT-193	Paper Receipt Claim to Voucher Unlock end-to-end	Commuter at 9 rides	1. Conductor records cash fare and prints receipt 2. Commuter scans receipt QR in mobile app	Commuter cycle reaches 10/10. Voucher wallet unlocks free ride voucher immediately.	
TC-INT-194	Commuter redeems Voucher on Conductor Terminal	Commuter holds voucher	1. Commuter shows voucher QR 2. Conductor scans voucher	Conductor terminal records free ride. Commuter voucher moves to "Used Vouchers" in real time.	
TC-INT-195	Admin modifies Fare Matrix on Web	Admin updates fare on web portal	1. Admin changes base fare from PHP 15 to PHP 16 and publishes	Both Commuter and Conductor mobile apps fetch updated fare matrix on next calculation.	
TC-INT-196	Admin publishes Route Detour on Web	Road construction advisory	1. Admin publishes temporary route detour on web	Both mobile apps render the updated route geometry and detour advisory banner.	
TC-INT-197	Admin publishes System Announcement	Emergency or promo notice	1. Admin publishes announcement	Commuter mobile bottom nav badge updates immediately. Announcement appears in feed.	
TC-INT-198	Lost & Found Claim workflow end-to-end	Conductor logs found umbrella	1. Item posted in catalog 2. Commuter files claim with proof 3. Admin approves claim on web	Commuter app updates claim status to "Ready for Release" with pickup instructions.	
TC-INT-199	Commuter Feedback reflects on Conductor metrics	Ride completed	1. Commuter rates conductor 5 stars with "Polite" tag	Conductor's "Performance Metrics" screen updates average ratings and positive tag counts.	
TC-INT-200	Conductor SOS reflects on Admin Live Monitoring	Conductor triggers SOS on phone	1. Tap SOS on mobile conductor app	Admin web dashboard flashes red siren banner with vehicle unit plate, coordinates, and driver details.	
TC-INT-201	Concurrent Web and Mobile Conductor sessions	Conductor opens web portal and mobile app	1. Perform shift check on mobile 2. Open web portal on PC	Mobile operates shift; web acts as spectator or management without kicking mobile session.	
TC-INT-202	Concurrent Web and Mobile Commuter sessions	Commuter signed in on PC and phone	1. Check lost items on PC 2. Hail jeepney on phone	Neither session is dropped. Both receive account updates.	

B. Mobile Hardware & Offline Resilience

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-SYS-203	Camera permission denial and recovery	Fresh app install	1. Deny camera permission 2. Tap "Pay with GCash" 3. Tap "Open Settings" 4. Enable camera permission in phone settings and return	Viewfinder initializes immediately upon returning to app without needing app restart.	
TC-SYS-204	Camera flashlight / torch toggle	QR scanner open in dark environment	1. Tap torch button in scanner	Device flashlight turns on to illuminate ticket/QR code. Tapping again turns torch off.	

Test Case ID	Test Case Description	Pre-conditions	Steps	Expected Result	Status
TC-SYS-205	Rapid barcode autofocus	Blurry receipt barcode	1. Move camera closer/further from thermal receipt	Camera autofocuses rapidly (<500ms) and decodes token reliably.	
TC-SYS-206	Background location tracking persistence	Conductor actively driving unit	1. Send conductor app to background (switch to another app or home screen) for 15 minutes	Vehicle continues transmitting GPS points periodically. Admin map tracking remains continuous.	
TC-SYS-207	GPS loss handling in tunnel or dead zone	Transit enters area with no GPS signal	1. Simulate GPS loss / airplane GPS mode	App gracefully retains last known coordinate. Displays "GPS signal weak" banner without crashing. Resumes smoothly when signal returns.	
TC-SYS-208	Complete offline storage integrity (AsyncStorage)	5 cash fares stored offline	1. Force kill the app while offline 2. Reopen the app while still offline	All 5 queued offline fares remain intact in local storage. Shift cash total accurately reflects the un-synced fares.	
TC-SYS-209	Network drop during fare calculation	Commuter calculating fare	1. Turn off WiFi / cellular data during stop selection	App uses locally cached Fare Matrix and route data. Displays fare calculation without network error.	
TC-SYS-210	App killed and restored during active shift	Conductor has active shift	1. Force kill **chatco-mobile** 2. Relaunch app	App reads stored session token and shift ID, re-hydrates dashboard and operating device status seamlessly.	
TC-SYS-211	Deep-linking from GCash payment return	Commuter completed payment in external GCash app	1. GCash app triggers return URL **chatco-commuter://gcash/return?status=paid**	App foregrounds automatically, parses return parameters, and renders Payment Success receipt screen.	
TC-SYS-212	Memory leak and stability on long shift	Conductor app active for 8+ hours	1. Operate app continuously across 100+ transactions	Metro memory consumption remains stable (<180MB RAM). No UI stutter or memory crashes.	
TC-SYS-213	High-contrast dark theme outdoor readability	Device in bright sunlight	1. Test app under direct sunlight	Dark theme (**#050F1A** background with **#62A0EA** and **#FF6D3A** high-contrast accents) provides clear legibility for drivers and conductors.	
TC-SYS-214	Audio chime and haptic feedback toggle	Conductor Settings open	1. Toggle "Scan Sound" off 2. Scan barcode	Barcode scans silently without audio beep, respecting user preference.	

Notes: possible issues found while reading the code

Notes 1 to 5 are bugs: those test cases are expected to be marked F until fixed. Notes 6 to 8 are observations: the linked cases should pass, but the behavior may need a decision.

No code has been changed for any of these.

1. TC-COMM-AUTH-015: Uploading a non-image file (e.g. PDF, Word) displays an error, but the camera capture button allows capturing photos without validating orientation or minimum dimensions.

2. TC-COMM-HAIL-025: When commuter is outside the 1km radius, the hail button is disabled ('TOO FAR'), but if GPS drifts across the 1km boundary momentarily, a fast tap can dispatch an invalid hail before the lockout updates.

3. TC-COND-DEV-123: Secondary device logins enter View-Only mode as expected, but active shift polling periodically prompts an unnecessary re-render of the vehicle selector modal.

4. TC-COND-REMIT-182: Remittance submission blocks unsynced cash transactions, but the alert modal does not enumerate which specific trip IDs remain in the queue.

5. TC-COMM-REWARD-053: Paper cash receipts older than 7 days fail claim validation, but the error message states 'Invalid or expired receipt' without specifying whether the token was mistyped or past expiration.

6. TC-COMM-AUTH-017: Platform-scoped Sanctum tokens allow concurrent mobile and web sessions without token drop; verify with backend team whether single-session lockout should ever apply across disparate platforms.

7. TC-SYS-RELIAB-209: Offline cash transactions generate client-side UUID tokens and queue locally; if the app cache is explicitly cleared before reconnecting to cellular data, un-synced cash tickets cannot be recovered.

8. TC-COND-STAT-133: Background GPS broadcasting relies on Expo Task Manager; aggressive Android battery saver policies (e.g., Xiaomi MIUI, Samsung OneUI) may throttle location updates after 15 minutes of screen-off idling.